Question 5: Deploy an application to a Kubernetes (Minikube) cluster using Deployment and Service manifests, and verify the running pods.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. Kubernetes manifests

k8s/configmap.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: lab2-config
data:
  GREETING: "Hello from Meenakshi Kubernetes Lab"
```

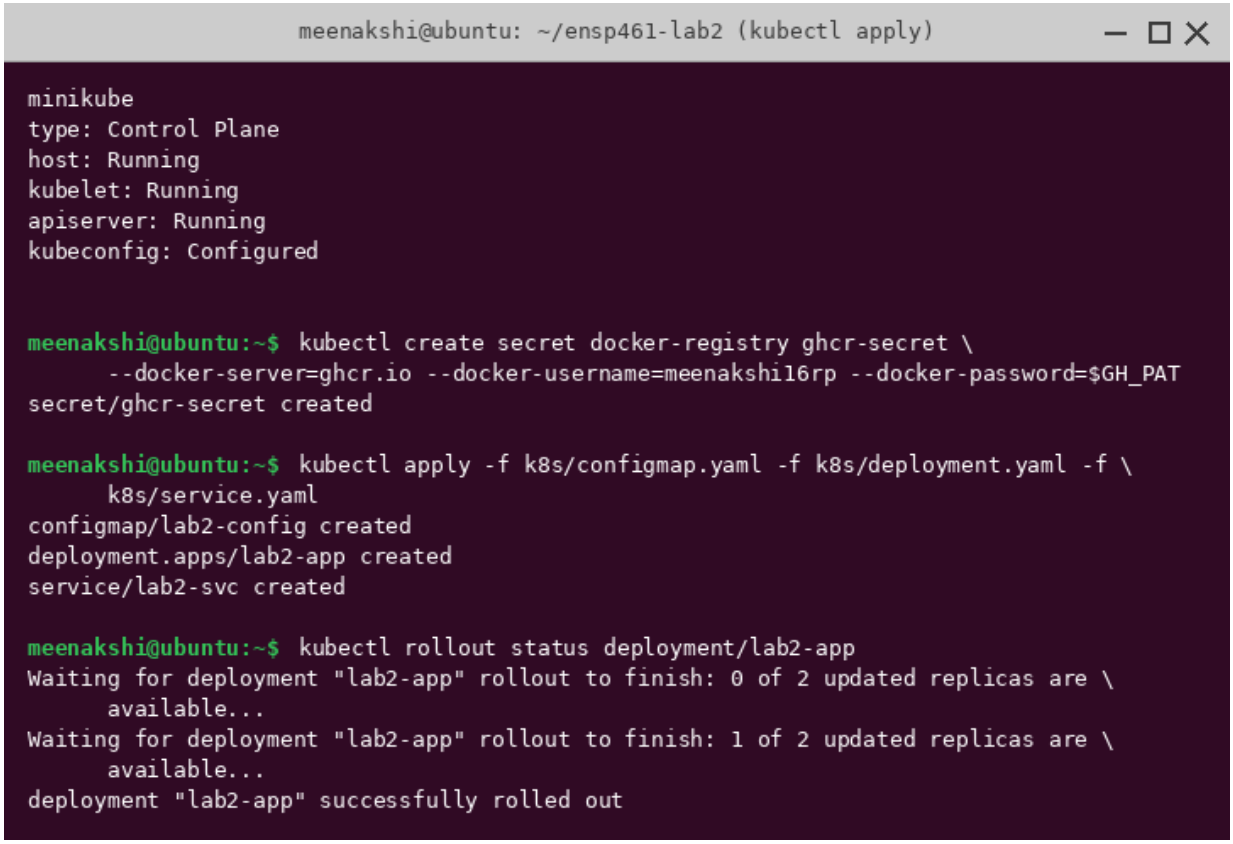
k8s/deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lab2-app
  labels:
    app: lab2-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: lab2-app
  template:
    metadata:
      labels:
        app: lab2-app
    spec:
      imagePullSecrets:
        - name: ghcr-secret
      containers:
        - name: lab2-app
          image: ghcr.io/meenakshi16rp/devops-lab2-app:2.0.0
          ports:
            - containerPort: 5000
          env:
            - name: GREETING
              valueFrom:
                configMapKeyRef:
                  name: lab2-config
                  key: GREETING
```

k8s/service.yaml

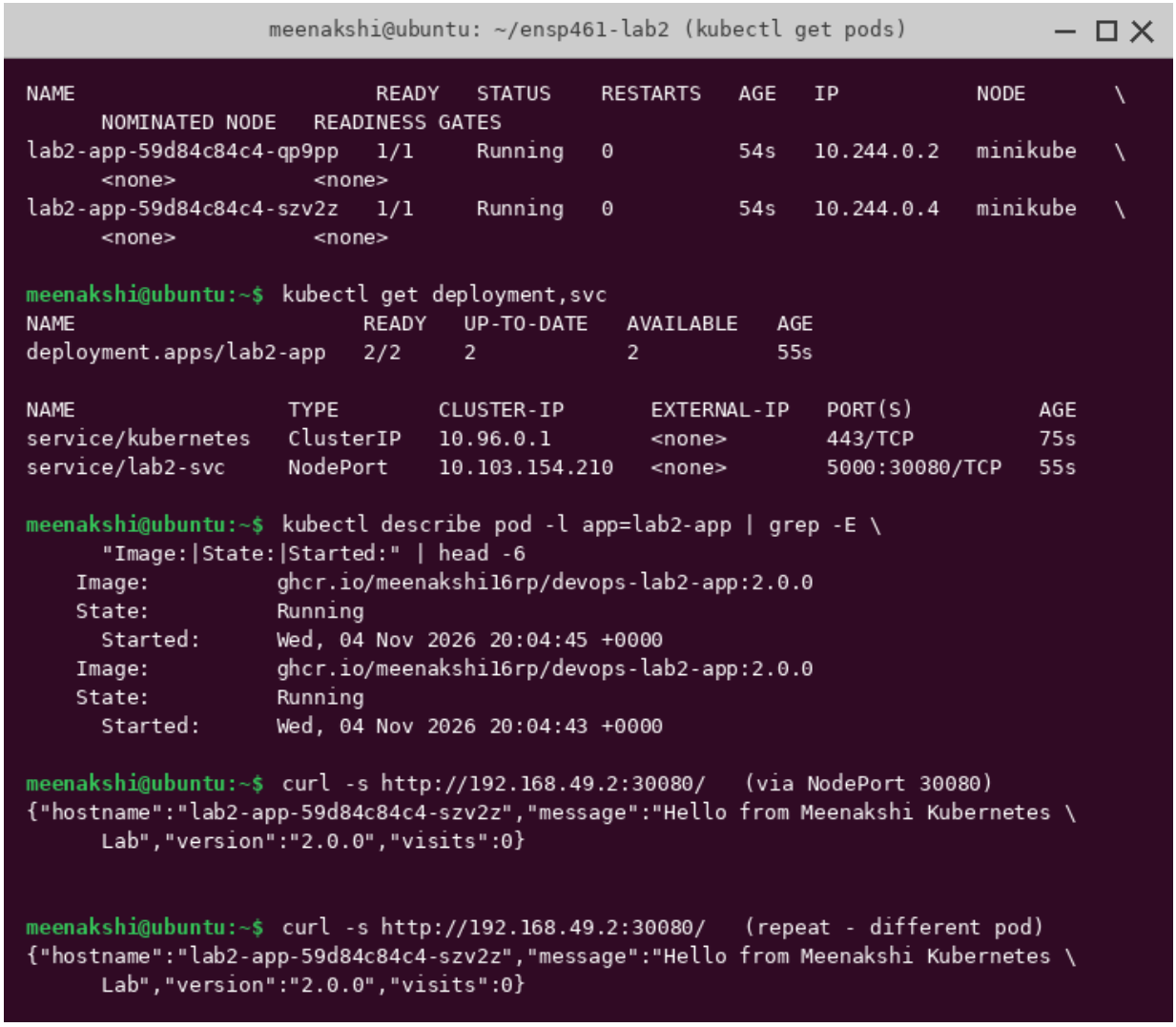
```
apiVersion: v1
kind: Service
metadata:
  name: lab2-svc
spec:
  type: NodePort
  selector:
    app: lab2-app
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30080
```

2. Starting Minikube and deploying



- minikube start --driver=docker brings up a single-node Kubernetes v1.37.0 cluster in a Docker container.
- A docker-registry secret (ghcr-secret) authorizes the kubelet to pull the private image from ghcr.io; the Deployment references it via imagePullSecrets.
- The GREETING environment variable is injected from the ConfigMap.

3. Verifying the running pods and the service



4. Observations

- Both replicas reached Running 1/1 and the NodePort service (30080) served the application at http://192.168.49.2:30080 with the ConfigMap greeting.
- kubectl rollout status is the reliable way to wait for availability instead of polling manually.
- The first minikube start hit an apiserver timeout (binary download hiccup inside the node); minikube delete followed by a clean start resolved it.